

Bio-Sense-Us with Biosensors



Akul Sabhanwal is an electronics and communication engineer. He is currently working with Simmtronics Semiconductors Ltd

A biosensor is an analytical device that converts a biological response into an electrical signal. The term biosensor is often used for sensor devices used to determine the concentration of substances and other parameters of biological interest even where these do not utilise a biological system directly.

With an estimated 60 per cent annual growth rate, the major demand for biosensors is coming from health-care industry, but with some pressure from other areas such as food-quality appraisal and environmental monitoring. The current types are potentiometric and amperometric biosensors and colourimetric paper enzyme strips. However, all main transducer types are likely to be thoroughly examined, for use in biosensors, over the next few years.

A successful biosensor must possess at least some of the following features:

1. The biocatalyst must be highly specific for the purpose of analyses, be stable under normal storage conditions and show good stability over a large number of assays.
2. The reaction should be as independent of such physical parameters as stirring, pH and temperature as is manageable.
3. The response should be accurate, precise, reproducible and linear over the useful analytical range, without dilution or concentration. It should also be free from electrical noise.

4. The complete biosensor should be cheap, small, portable and capable of being used by semi-skilled people.

Besides, there should be a market for the biosensor. There is little purpose developing one if other factors encourage the use of traditional methods and discourage decentralisation of laboratory testing.

How does it work

The key part of a biosensor is the transducer (shown as the green box in Fig. 1), which makes use of a physical change accompanying the reaction, which may be:

1. The heat output (or absorbed) by the reaction (calorimetric biosensors)
2. Changes in distribution of charges causing an electrical potential to be produced (potentiometric biosensors)
3. Movement of electrons produced in a redox reaction (amperometric biosensors)
4. Light output during the reaction or a light absorbance difference between reactants and products (optical biosensors)
5. Effects due to the mass of reactants or products (piezoelectric biosensors)

There are three so-called generations of biosensors. First-generation biosensors are those in which the normal product of the reaction diffuses to the transducer and causes an electrical response. Second-generation biosensors involve specific mediators between the reaction and the transducer in order to generate improved response. And third-generation biosensors are those in which the reaction itself causes the response and no product or mediator diffusion is directly involved.

An electrical signal from the transducer is often low and superimposed upon a relatively high and noisy (that is, containing a high-frequency signal component of an apparently random nature, due to electrical interference or generated within the electronic components of

Fig. 1: Schematic diagram showing the main components of a biosensor. The biocatalyst (a) converts the substrate to product. This reaction is determined by the transducer (b), which converts it to an electrical signal. The output from the transducer is amplified (c), processed (d) and displayed (e)

